Cell-Type Annotation

2026-04-17

Introduction

After clustering, each cluster must be labelled with a biological identity: T cell, B cell, macrophage, tumour epithelial cell, neuron subtype. Cell-type annotation is the bridge between unsupervised clustering and biological interpretation. Two main approaches compete: reference-based methods (SingleR, Azimuth, scType) that score each cell or cluster against curated reference profiles, and manual annotation that inspects marker genes against literature and reference atlases.

Prerequisites

A working understanding of clustered scRNA-seq output, marker-gene analysis, and the available cell-type reference atlases (Human Cell Atlas, Tabula Muris, Monaco Immune, ENCODE).

Theory

SingleR computes per-cell Spearman correlations against each reference profile over reference-defined marker genes; cells are labelled with the highest-scoring reference label. A pruning step removes calls below a confidence threshold. The method is fast, automated, and produces both per-cell and per-cluster summaries.

Azimuth uses a transfer-learning approach: it projects query cells into the reference's latent space (typically a PCA + harmony-corrected representation) and transfers labels via nearest-neighbour matching. It is the modern Seurat-team-recommended approach when a curated reference atlas exists for the tissue.

Manual annotation inspects each cluster's top marker genes against curated marker lists (canonical T-cell markers: CD3D, CD3E; B cells: CD19, MS4A1; macrophages: CD68, CD163). Cluster-by-cluster manual annotation is essential when references are missing or the tissue is non-standard.

Assumptions

The reference atlas matches the tissue and species of interest; cell types present in the data are represented in the reference (rare or unusual states may be absent and require manual annotation).

R Implementation

```r
library(SingleR); library(celldex)

# Reference: built-in Monaco immune cells
ref <- MonacoImmuneData()

# Toy query: simulate a cell-by-gene expression matrix
set.seed(2026)
query <- matrix(rpois(200 * 100, lambda = 2), 200, 100)
rownames(query) <- sample(rownames(ref), 200)   # match gene names
colnames(query) <- paste0("c", 1:100)

pred <- SingleR(test = query, ref = ref,
```

```r
                 labels = ref$label.main,
                 de.method = "wilcox")

table(pred$labels)
head(pred[, c("labels", "pruned.labels")])
```

Output & Results

`SingleR()` returns per-cell predicted labels, pruned labels (high-confidence subset), and the underlying per-reference-label scores. Summary tables show predicted cell-type counts; `pruneScores()` allows fine-tuning of the confidence threshold.

Interpretation

A reporting sentence: "SingleR with the Monaco Immune reference assigned 62 % of cells to canonical PBMC types (T cells, B cells, NK cells, monocytes, dendritic cells); pruning removed 8 % low-confidence calls. The remaining 30 % unannotated cells were tumour-associated stromal cells confirmed manually using ACTA2, COL1A1, and CD90 markers." Always report the reference, pruning rate, and manual validation of automated calls.

Practical Tips

- Use a reference matching the tissue source — never annotate skin cells against a blood reference; the cell types simply do not match.
- Report pruned labels rather than raw labels; pruned labels exclude low-confidence calls that often reflect intermediate or unusual states.
- For novel tissues or rare states, manual annotation outperforms reference-based methods; the references cannot label what they have not seen.
- Azimuth (`SeuratDisk::LoadH5Seurat` + `MapQuery`) is the modern Seurat alternative; it handles batch differences between query and reference more gracefully than SingleR.
- Cross-validate automated cluster labels against marker gene expression; a cluster labelled "T cell" by SingleR but without CD3D/CD3E expression is suspect and warrants investigation.
- For very granular reference atlases (Human Cell Atlas with 100+ cell types), use cluster-level SingleR rather than per-cell to reduce noise.

R Packages Used

`SingleR` for the canonical reference-based annotation; `celldex` for built-in reference datasets (Monaco, Encode, Hematopoietic Cell Atlas, etc.); `Azimuth` for Seurat-flavoured transfer learning; `scType` for marker-database-based annotation; `clustifyr` for an alternative reference-based approach.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)

- scRNA-seq with Seurat